

DTSA 5507

Fundamentals of Software Architecture for Big Data

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (100.0 / 100)	83.3 / 100
SPECIALIZATION Software Development Lifecycle for Data Scientists		

83

CURRENT AVG SCORE
across 6 job roles

100

POTENTIAL SCORE
with recommended additions

+32

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Intended for individuals looking to understand the basics of software engineering as they relate to building large software systems that leverage big data. Students will be introduced to software engineering concepts necessary to build and scale large, data intensive, distributed systems. Starting with software engineering best practices and loosely coupled, highly cohesive data microservices, the course takes students through the evolution of a distributed system over time. Formerly offered as a special topics course.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	100.0	100.0	+0.0	LOW
ML Engineer	89.4	100.0	+10.6	LOW
Machine Learning Engineer	88.5	100.0	+11.5	LOW
MLOps Engineer	84.9	100.0	+15.1	MEDIUM
Applied Scientist	68.8	100.0	+31.2	HIGH
Data Scientist	68.2	100.0	+31.8	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Business Communication

2 of 6 roles

insights

business

LLMs & Generative AI

2 of 6 roles

storage

Model Deployment & Serving

2 of 6 roles

production

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

MEDIUM

Consider Business Communication content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.

2

MEDIUM

Consider LLMs & Generative AI content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: storage.

3

MEDIUM

Consider Model Deployment & Serving content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.

GAP KEYWORDS BY ROLE

ML Engineer

89.4 → 100.0 (+10.6 pts)

discovery

learning

machine

models

production

infrastructure

world

problems

Machine Learning Engineer

88.5 → 100.0 (+11.5 pts)

learning

analytics

tools

solutions

business

mobile

design

science

MLOps Engineer

84.9 → 100.0 (+15.1 pts)

infrastructure

production

learning

machine

tools

models

Applied Scientist

68.8 → 100.0 (+31.2 pts)

discovery

scientific

research

storage

problems

models

technical

learning

Data Scientist

68.2 → 100.0 (+31.8 pts)

toolkit

product

characteristics

analytics

scripting

insights

social

problems

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.